T3 E_-2: general EFT1 RGEs

Matchete/RGBeta symbolic output in mathematical notation

E_-2

Additional quartics: 6, additional bilinears: 1, extra Yukawa: y2Conj.
General matching and Wilson coefficients are not computed in this symbolic report.

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{47{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{11{g_2}^3}{6}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = & \, 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = & \, \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} &= 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ &+ 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \frac{3}{2}\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij} \\ &- \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ &- \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

m_1^2

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} &= 4\mathrm{muT3GeneralS2Sq}(3\,(\mathrm{lambdaT3GeneralQ001102I1})^* + (\mathrm{lambdaT3GeneralQ001102I2})^*) \\ &+ 12\mathrm{lambdaT3GeneralQ001102I1}\,(\mathrm{muT3GeneralS2Sq})^* \\ &+ 4\mathrm{lambdaT3GeneralQ001102I2}\,(\mathrm{muT3GeneralS2Sq})^* \\ &- 12{g_2}^2 m_1^2 \\ &- 6{g_1}^2 m_1^2 \\ &+ m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\ &+ m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})\end{aligned}$$

m_2^2

$$\begin{aligned}16\pi^2 \frac{dm_2^2}{d\ln\mu} &= 20\mathrm{muT3GeneralS2Sq}\,(\mathrm{lambdaT3GeneralQ000013I1})^* \\ &+ 20\mathrm{lambdaT3GeneralQ000013I1}\,(\mathrm{muT3GeneralS2Sq})^* \\ &- 12{g_2}^2 m_2^2 \\ &+ m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\ &+ m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})\end{aligned}$$

λ_1

$$\begin{aligned}16\pi^2 \frac{d\lambda_1}{d\ln\mu} &= \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\ &- 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big) \\ &- 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\ &- 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big) \\ &+ 4\lambda_5\,(\lambda_5)^* \\ &+ 4\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^* \\ &+ 24\mathrm{lambdaT3GeneralQ110002I1}\,(\mathrm{lambdaT3GeneralQ110002I1})^* \\ &+ \frac{9{g_2}^4}{4}\end{aligned}$$

$$
\begin{aligned}
&+ {g_1}^2 \left(\frac{3{g_2}^2}{2} - 3\lambda_1 \right) \\
&- 9{g_2}^2\lambda_1 \\
&+ \frac{3{g_1}^4}{4} \\
&+ 12{\lambda_1}^2 \\
&+ 6{\lambda_{HS_1}^{(1)}}^2 \\
&+ 3{\lambda_{HS_1}^{(A)}}^2 \\
&+ 6{\lambda_{HS_2}^{(1)}}^2 \\
&+ 3{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = {} & 8\text{lambdaT3GeneralQ001102I1}(3\left(\text{lambdaT3GeneralQ001102I1}\right)^* + \left(\text{lambdaT3GeneralQ001102I2}\right)^*) \\
& + \text{lambdaT3GeneralQ001102I2}\left(8\left(\text{lambdaT3GeneralQ001102I1}\right)^* + \frac{20\left(\text{lambdaT3GeneralQ001102I2}\right)^*}{3}\right) \\
& + 22{g_2}^4 \\
& + {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_1}^{(1)}\right) \\
& - 24{g_2}^2\lambda_{S_1}^{(1)} \\
& + 12{g_1}^4 \\
& + 6{\lambda_{12}^{(1)}}^2 \\
& + 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
& + \frac{20{\lambda_{12}^{(\times)}}^2}{3} \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + 14{\lambda_{S_1}^{(1)}}^2 \\
& + 8\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
& + 16{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = {} & \frac{832}{3}\text{lambdaT3GeneralQ000004I1}\left(\text{lambdaT3GeneralQ000004I1}\right)^* \\
& + \frac{352}{3}\text{lambdaT3GeneralQ000013I1}\left(\text{lambdaT3GeneralQ000013I1}\right)^* \\
& + \frac{8}{3}\text{lambdaT3GeneralQ001102I1}(3\left(\text{lambdaT3GeneralQ001102I1}\right)^* + \left(\text{lambdaT3GeneralQ001102I2}\right)^*) \\
& + \text{lambdaT3GeneralQ001102I2}\left(\frac{8\left(\text{lambdaT3GeneralQ001102I1}\right)^*}{3} + 4\left(\text{lambdaT3GeneralQ001102I2}\right)^*\right) \\
& + \frac{16}{3}\text{lambdaT3GeneralQ110002I1}\left(\text{lambdaT3GeneralQ110002I1}\right)^* \\
& + 22{g_2}^4 \\
& - 24{g_2}^2\lambda_{S_2}^{(1)}
\end{aligned}
$$

$$
\begin{aligned}
&+ 6{\lambda_{12}^{(1)}}^2 \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
&+ \frac{20{\lambda_{12}^{(\times)}}^2}{3} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 14{\lambda_{S_2}^{(1)}}^2 \\
&+ 8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 16{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_1}^{(1)}(6\,\mathrm{Tr}\Big(Y_d\ {(Y_d)}^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ {(Y_e)}^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ {(Y_u)}^\dagger\Big) + 4\lambda_{S_1}^{(A)}) \\
& + 4\lambda_5\,{(\lambda_5)}^* \\
& + 4\mathrm{lambdaT3GeneralQ110002I1}(3\,{(\mathrm{lambdaT3GeneralQ001102I1})}^* + {(\mathrm{lambdaT3GeneralQ001102I2})}^*) \\
& + 12\mathrm{lambdaT3GeneralQ001102I1}\,{(\mathrm{lambdaT3GeneralQ110002I1})}^* \\
& + 4\mathrm{lambdaT3GeneralQ001102I2}\,{(\mathrm{lambdaT3GeneralQ110002I1})}^* \\
& + 4\mathrm{lambdaT3GeneralQ020101I1}\,{(\mathrm{lambdaT3GeneralQ020101I1})}^* \\
& + 6{g_2}^4 \\
& - \frac{33{g_2}^2\lambda_{HS_1}^{(1)}}{2} \\
& + 3{g_1}^4 \\
& - \frac{15{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
& + \lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
& + 6\lambda_1\lambda_{HS_1}^{(1)} \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + 8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
& + 6{\lambda_{HS_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_2}^{(1)}(6\,\mathrm{Tr}\Big(Y_d\ {(Y_d)}^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ {(Y_e)}^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ {(Y_u)}^\dagger\Big) + 4\lambda_{S_2}^{(A)}) \\
& + 4\lambda_5\,{(\lambda_5)}^* \\
& + 20\mathrm{lambdaT3GeneralQ000013I1}\,{(\mathrm{lambdaT3GeneralQ110002I1})}^* \\
& + 4\mathrm{lambdaT3GeneralQ110002I1}(5\,{(\mathrm{lambdaT3GeneralQ000013I1})}^* + 4\,{(\mathrm{lambdaT3GeneralQ110002I1})}^*) \\
& + 4\mathrm{lambdaT3GeneralQ020101I1}\,{(\mathrm{lambdaT3GeneralQ020101I1})}^* \\
& + 6{g_2}^4 \\
& - \frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2} \\
& - \frac{3{g_1}^2\lambda_{HS_2}^{(1)}}{2}
\end{aligned}
$$

$$
\begin{aligned}
&+ \lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ 6{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{12}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} &= 2\lambda_5 \left(\lambda_5\right)^* \\
&+ 4\text{lambdaT3GeneralQ000013I1}(5\left(\text{lambdaT3GeneralQ001102I1}\right)^* + \left(\text{lambdaT3GeneralQ001102I2}\right)^*) \\
&+ 4\text{lambdaT3GeneralQ001102I1}(5\left(\text{lambdaT3GeneralQ000013I1}\right)^* + 4\left(\text{lambdaT3GeneralQ001102I1}\right)^*) \\
&+ 4\text{lambdaT3GeneralQ001102I2}((\text{lambdaT3GeneralQ000013I1})^* + \left(\text{lambdaT3GeneralQ001102I2}\right)^*) \\
&+ 2\text{lambdaT3GeneralQ020101I1}\left(\text{lambdaT3GeneralQ020101I1}\right)^* \\
&+ \frac{63{g_2}^4}{5} \\
&- 24{g_2}^2\lambda_{12}^{(1)} \\
&- 6{g_1}^2\lambda_{12}^{(1)} \\
&+ 4{\lambda_{12}^{(1)}}^2 \\
&+ \lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
&+ \lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)}) \\
&+ \lambda_{12}^{(1)}(4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)}) \\
&+ 16{\lambda_{12}^{(A)}}^2 \\
&+ 4{\lambda_{12}^{(\times)}}^2 \\
&+ 4\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
&+ 4\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)} \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_5}{d\ln\mu} &= \lambda_5 \left(6\,\mathrm{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_{HS_2}^{(A)}}{3}\right) \\
&+ 4\text{lambdaT3GeneralQ001102I1}\left(\text{lambdaT3GeneralQ020101I1}\right)^* \\
&- 2\text{lambdaT3GeneralQ001102I2}\left(\text{lambdaT3GeneralQ020101I1}\right)^* \\
&+ 8\text{lambdaT3GeneralQ110002I1}\left(\text{lambdaT3GeneralQ020101I1}\right)^* \\
&- \frac{33{g_2}^2\lambda_5}{2} \\
&- \frac{9{g_1}^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_1}^{(A)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{HS_1}^{(A)}}{d\ln\mu} = \lambda_{HS_1}^{(A)}(&6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_1}^{(A)})\\ &- \frac{4}{3}\lambda_5\,(\lambda_5)^*\\ &- \frac{4}{3}\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^*\\ &+ {g_1}^2\left(-4{g_2}^2 - \frac{15\lambda_{HS_1}^{(A)}}{2}\right)\\ &- \frac{33{g_2}^2\lambda_{HS_1}^{(A)}}{2}\\ &+ 12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}\\ &+ 2\lambda_1\lambda_{HS_1}^{(A)}\\ &+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}\\ &+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}\end{aligned}$$

$\lambda_{HS_2}^{(A)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = \lambda_{HS_2}^{(A)}(&6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_2}^{(A)})\\ &- \frac{8}{9}\lambda_5\,(\lambda_5)^*\\ &- \frac{4}{3}\mathrm{lambdaT3GeneralQ020101I1}\,(\mathrm{lambdaT3GeneralQ020101I1})^*\\ &- \frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}\\ &- \frac{3{g_1}^2\lambda_{HS_2}^{(A)}}{2}\\ &+ 12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}\\ &+ 2\lambda_1\lambda_{HS_2}^{(A)}\\ &+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}\\ &+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}\end{aligned}$$

$\lambda_{S_1}^{(A)}$

$$\begin{aligned}16\pi^2 \frac{d\lambda_{S_1}^{(A)}}{d\ln\mu} = &-\frac{4}{3}\mathrm{lambdaT3GeneralQ001102I2}\,(\mathrm{lambdaT3GeneralQ001102I2})^*\\ &- 2{g_2}^4\\ &+ {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_1}^{(A)}\right)\\ &- 24{g_2}^2\lambda_{S_1}^{(A)}\\ &+ 12{\lambda_{12}^{(A)}}^2\\ &- \frac{4{\lambda_{12}^{(\times)}}^2}{3}\end{aligned}$$

$$
\begin{aligned}
&+ 3{\lambda_{HS_1}^{(A)}}^2 \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)} \\
&+ 20{\lambda_{S_1}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(A)}}{d\ln\mu} = &-\frac{224}{3}\text{lambdaT3GeneralQ000004I1}\,(\text{lambdaT3GeneralQ000004I1})^* \\
&- \frac{44}{3}\text{lambdaT3GeneralQ000013I1}\,(\text{lambdaT3GeneralQ000013I1})^* \\
&- \frac{4}{3}\text{lambdaT3GeneralQ001102I2}\,(\text{lambdaT3GeneralQ001102I1})^* \\
&- \frac{4}{3}\text{lambdaT3GeneralQ001102I1}(3\,(\text{lambdaT3GeneralQ001102I1})^* + (\text{lambdaT3GeneralQ001102I2})^*) \\
&- \frac{8}{3}\text{lambdaT3GeneralQ110002I1}\,(\text{lambdaT3GeneralQ110002I1})^* \\
&- 2{g_2}^4 \\
&- 24{g_2}^2\lambda_{S_2}^{(A)} \\
&+ 12{\lambda_{12}^{(A)}}^2 \\
&- \frac{4{\lambda_{12}^{(\times)}}^2}{3} \\
&+ 3{\lambda_{HS_2}^{(A)}}^2 \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{12}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(A)}}{d\ln\mu} = &-\frac{2}{9}\lambda_5\,(\lambda_5)^* \\
&- \frac{1}{3}\text{lambdaT3GeneralQ020101I1}\,(\text{lambdaT3GeneralQ020101I1})^* \\
&- 24{g_2}^2\lambda_{12}^{(A)} \\
&- 6{g_1}^2\lambda_{12}^{(A)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\
&+ \frac{16\lambda_{12}^{(A)}\lambda_{12}^{(\times)}}{3} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ 16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)} \\
&+ 3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}
\end{aligned}
$$

$\lambda_{12}^{(\times)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(\times)}}{d\ln\mu} = & -\lambda_5 \left(\lambda_5\right)^* \\
& + \text{lambdaT3GeneralQ001102I2}(4 \left(\text{lambdaT3GeneralQ000013I1}\right)^* + 8 \left(\text{lambdaT3GeneralQ001102I1}\right)^* + 10 \left(\text{lam} \\
& + 4\text{lambdaT3GeneralQ000013I1} \left(\text{lambdaT3GeneralQ001102I2}\right)^* \\
& + 8\text{lambdaT3GeneralQ001102I1} \left(\text{lambdaT3GeneralQ001102I2}\right)^* \\
& - \text{lambdaT3GeneralQ020101I1} \left(\text{lambdaT3GeneralQ020101I1}\right)^* \\
& + \frac{51{g_2}^4}{10} \\
& - 24{g_2}^2\lambda_{12}^{(\times)} \\
& - 6{g_1}^2\lambda_{12}^{(\times)} \\
& + 8\lambda_{12}^{(1)}\lambda_{12}^{(\times)} \\
& + 10{\lambda_{12}^{(\times)}}^2 \\
& + 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)} \\
& - 2\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)} \\
& + 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)} \\
& - 2\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}
\end{aligned}
$$

$lambdaT3GeneralQ000004I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ000004I1}{d\ln\mu} = & -24{g_2}^2\text{lambdaT3GeneralQ000004I1} \\
& + 12\lambda_{S_2}^{(1)}\text{lambdaT3GeneralQ000004I1} \\
& - 8\lambda_{S_2}^{(A)}\text{lambdaT3GeneralQ000004I1} \\
& + 11\text{lambdaT3GeneralQ000013I1}^2 \\
& + 3\text{lambdaT3GeneralQ001102I1}^2 \\
& + 2\text{lambdaT3GeneralQ001102I1lambdaT3GeneralQ001102I2} \\
& + \text{lambdaT3GeneralQ001102I2}^2 \\
& + 2\text{lambdaT3GeneralQ110002I1}^2
\end{aligned}
$$

$lambdaT3GeneralQ000013I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ000013I1}{d\ln\mu} = & 88\text{lambdaT3GeneralQ000004I1} \left(\text{lambdaT3GeneralQ000013I1}\right)^* \\
& - 24{g_2}^2\text{lambdaT3GeneralQ000013I1} \\
& + \lambda_{12}^{(1)}(6\text{lambdaT3GeneralQ001102I1} + 2\text{lambdaT3GeneralQ001102I2}) \\
& + \lambda_{12}^{(\times)}(4\text{lambdaT3GeneralQ001102I1} + 4\text{lambdaT3GeneralQ001102I2}) \\
& + 4\lambda_{HS_2}^{(1)}\text{lambdaT3GeneralQ110002I1} \\
& + 22\lambda_{S_2}^{(1)}\text{lambdaT3GeneralQ000013I1}
\end{aligned}
$$

$lambdaT3GeneralQ001102I1$

$$
16\pi^2 \frac{dlambdaT3GeneralQ001102I1}{d\ln\mu} = 8\text{lambdaT3GeneralQ000004I1}(5 \left(\text{lambdaT3GeneralQ001102I1}\right)^* + (\text{lambdaT3Genera}
$$

$$
\begin{aligned}
&- 24{g_2}^2\text{lambdaT3GeneralQ001102I1} \\
&- 6{g_1}^2\text{lambdaT3GeneralQ001102I1} \\
&+ \lambda_{12}^{(1)}(10\text{lambdaT3GeneralQ000013I1} + 8\text{lambdaT3GeneralQ001102I1}) \\
&+ \lambda_{12}^{(\times)}(4\text{lambdaT3GeneralQ000013I1} + 4\text{lambdaT3GeneralQ001102I2}) \\
&+ 4\lambda_{HS_1}^{(1)}\text{lambdaT3GeneralQ110002I1} \\
&+ \lambda_{S_1}^{(1)}(8\text{lambdaT3GeneralQ001102I1} + 2\text{lambdaT3GeneralQ001102I2}) \\
&+ \lambda_{S_1}^{(A)}(4\text{lambdaT3GeneralQ001102I1} + 2\text{lambdaT3GeneralQ001102I2}) \\
&+ 2\lambda_{S_2}^{(1)}\text{lambdaT3GeneralQ001102I1} \\
&+ \lambda_{S_2}^{(A)}(-4\text{lambdaT3GeneralQ001102I1} - 2\text{lambdaT3GeneralQ001102I2}) \\
&+ 2\lambda_5\text{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$lambdaT3GeneralQ001102I2$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ001102I2}{d\ln\mu} &= 16\text{lambdaT3GeneralQ000004I1}\left(\text{lambdaT3GeneralQ001102I2}\right)^* \\
&- 24{g_2}^2\text{lambdaT3GeneralQ001102I2} \\
&- 6{g_1}^2\text{lambdaT3GeneralQ001102I2} \\
&+ 8\lambda_{12}^{(1)}\text{lambdaT3GeneralQ001102I2} \\
&+ \lambda_{12}^{(\times)}(8\text{lambdaT3GeneralQ000013I1} + 16\text{lambdaT3GeneralQ001102I1} + 20\text{lambdaT} \\
&+ 2\lambda_{S_1}^{(1)}\text{lambdaT3GeneralQ001102I2} \\
&- 2\lambda_{S_1}^{(A)}\text{lambdaT3GeneralQ001102I2} \\
&+ 2\lambda_{S_2}^{(1)}\text{lambdaT3GeneralQ001102I2} \\
&+ 2\lambda_{S_2}^{(A)}\text{lambdaT3GeneralQ001102I2} \\
&- 2\lambda_5\text{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$lambdaT3GeneralQ020101I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ020101I1}{d\ln\mu} &= \text{lambdaT3GeneralQ020101I1}(6\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
&+ 4\text{lambdaT3GeneralQ001102I1}\left(\lambda_5\right)^* \\
&- 2\text{lambdaT3GeneralQ001102I2}\left(\lambda_5\right)^* \\
&+ 8\text{lambdaT3GeneralQ110002I1}\left(\lambda_5\right)^* \\
&- \frac{33{g_2}^2\text{lambdaT3GeneralQ020101I1}}{2} \\
&- \frac{9{g_1}^2\text{lambdaT3GeneralQ020101I1}}{2} \\
&+ 2\lambda_{12}^{(1)}\text{lambdaT3GeneralQ020101I1} \\
&- 2\lambda_{12}^{(A)}\text{lambdaT3GeneralQ020101I1} \\
&- 2\lambda_{12}^{(\times)}\text{lambdaT3GeneralQ020101I1} \\
&+ 2\lambda_1\text{lambdaT3GeneralQ020101I1} \\
&+ 4\lambda_{HS_1}^{(1)}\text{lambdaT3GeneralQ020101I1} \\
&- 2\lambda_{HS_1}^{(A)}\text{lambdaT3GeneralQ020101I1} \\
&+ 4\lambda_{HS_2}^{(1)}\text{lambdaT3GeneralQ020101I1} \\
&- 2\lambda_{HS_2}^{(A)}\text{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$lambdaT3GeneralQ110002I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ110002I1}{d\ln\mu} = {} & \text{lambdaT3GeneralQ110002I1}(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
& + 40\text{lambdaT3GeneralQ000004I1}\ (\text{lambdaT3GeneralQ110002I1})^* \\
& - \frac{33{g_2}^2\text{lambdaT3GeneralQ110002I1}}{2} \\
& - \frac{3{g_1}^2\text{lambdaT3GeneralQ110002I1}}{2} \\
& + 6\lambda_1\text{lambdaT3GeneralQ110002I1} \\
& + \lambda^{(1)}_{HS_1}(6\text{lambdaT3GeneralQ001102I1} + 2\text{lambdaT3GeneralQ001102I2}) \\
& + \lambda^{(1)}_{HS_2}(10\text{lambdaT3GeneralQ000013I1} + 8\text{lambdaT3GeneralQ110002I1}) \\
& + 2\lambda^{(1)}_{S_2}\text{lambdaT3GeneralQ110002I1} \\
& - 4\lambda^{(A)}_{S_2}\text{lambdaT3GeneralQ110002I1} \\
& + 4\lambda_5\text{lambdaT3GeneralQ020101I1}
\end{aligned}
$$

$muT3GeneralS2Sq$

$$
\begin{aligned}
16\pi^2 \frac{dmuT3GeneralS2Sq}{d\ln\mu} = {} & 40\text{lambdaT3GeneralQ000004I1}\ (\text{muT3GeneralS2Sq})^* \\
& - 12{g_2}^2\text{muT3GeneralS2Sq} \\
& + 2\lambda^{(1)}_{S_2}\text{muT3GeneralS2Sq} \\
& - 4\lambda^{(A)}_{S_2}\text{muT3GeneralS2Sq} \\
& + 10\text{lambdaT3GeneralQ000013I1}m_2^2 \\
& + m_1^2(6\text{lambdaT3GeneralQ001102I1} + 2\text{lambdaT3GeneralQ001102I2})
\end{aligned}
$$